[1]:
```python
import numpy as np          # calculations
import pandas as pd         # dataset handle
import matplotlib.pyplot as plt   # graphs
import seaborn as sns       # better graphs
from scipy import stats     # statistics functions
```

[2]:
```python
df = pd.read_csv("spread_locator_dataset - spread_locator_dataset.csv.csv")
# Show data
print(df.head())
```

```
                       transaction_id customer_id  transaction_amount  \
0  e98aa092-3770-4fdb-9502-5b5a6a244811    CUST2824             3821.34
1  11ba6918-dba0-41e5-96cf-f5a7b95f0103    CUST1409             2781.84
2  82b7654b-6eb7-4579-89a0-1a9edec0a7bb    CUST5506             4120.97
3  f7166574-f400-4d53-b526-0b11f6619ddf    CUST5012             6383.78
4  8632fe26-b507-4068-9c68-1b2fa04fecb3    CUST4657             2651.61

  transaction_date  transaction_count region transaction_status
0        1/26/2023                  3  North               Fail
1        1/28/2023                  0   East               Fail
2        1/28/2023                  0  South               Fail
3        1/18/2023                  2  South            Success
4         1/4/2023                  4  North            Success
```

[3]:
```python
# Check columns
print("Columns in dataset:")
print(df.columns)

# Check numeric columns
numeric_cols = df.select_dtypes(include=np.number).columns
print("Numeric Columns:", numeric_cols)
```

```
Columns in dataset:
Index(['transaction_id', 'customer_id', 'transaction_amount',
       'transaction_date', 'transaction_count', 'region',
       'transaction_status'],
      dtype='object')
Numeric Columns: Index(['transaction_amount', 'transaction_count'], dtype='object')
```

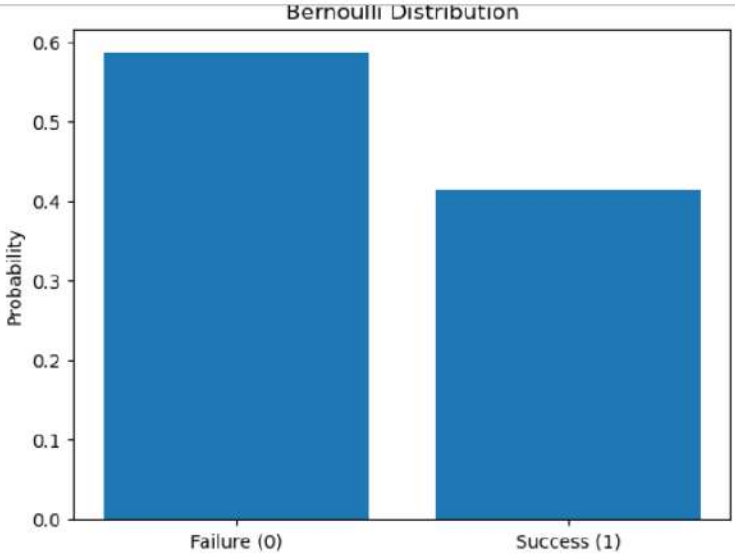
[4]:

```
amount = df[numeric_cols[0]]   # main data
data2 = df[numeric_cols[1]] if len(numeric_cols) > 1 else amount

transaction = (amount > amount.mean()).astype(int)
print("Data Prepared")
```

```
Data Prepared
```

[5]:

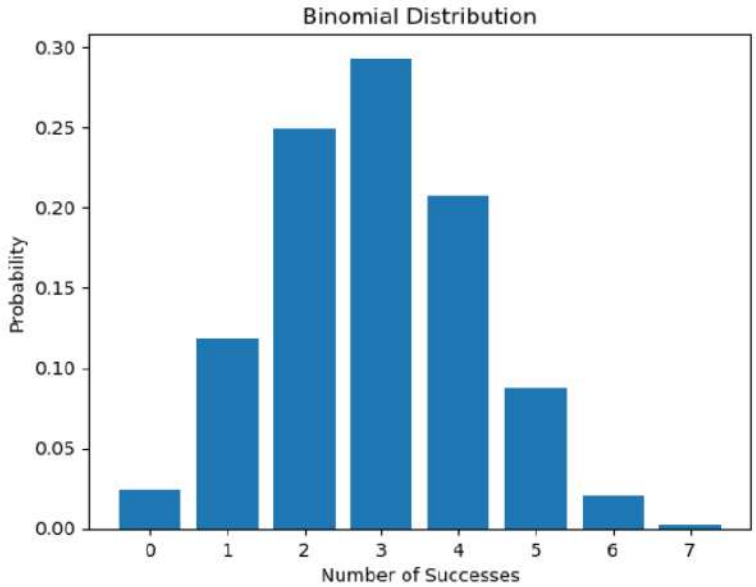
```
# Bernoulli Distribution

p = transaction.mean()

# Possible outcomes
x = [0, 1]
prob = [1 - p, p]

plt.bar(x, prob)
plt.xticks([0,1], ['Failure (0)', 'Success (1)'])
plt.title("Bernoulli Distribution")
plt.ylabel("Probability")
plt.show()

print("Explanation:")
print("The Bernoulli distribution has only 2 outcomes (0 or 1).")
print("This chart shows the probability of success (p) and failure (1-p).")
```




```
Explanation:
The Bernoulli distribution has only 2 outcomes (0 or 1).
This chart shows the probability of success (p) and failure (1-p).
```

[6]:
```
# Binomial Distribution

n = 7
x = np.arange(0, n+1)

binomial_probs = stats.binom.pmf(x, n, p)

plt.bar(x, binomial_probs)
plt.title("Binomial Distribution")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
```

```
plt.ylabel("Probability")
plt.show()

print("Explanation:")
print("The binomial distribution has multiple trials")
print("This chart shows the probability of successors from 0 to 7.")
```



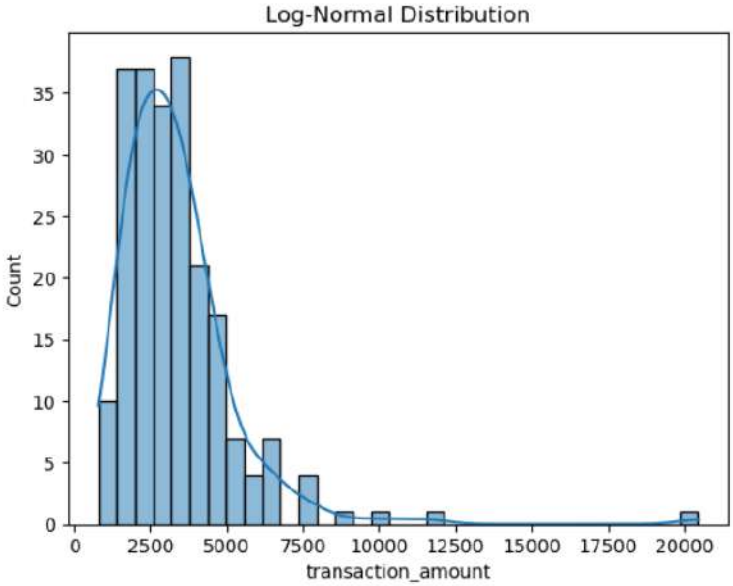

```
Explanation:
The binomial distribution has multiple trials
This chart shows the probability of successors from 0 to 7.
```

[7]:
```python
# Poisson

lam = data2.mean()
print("Lambda:", lam)
poisson_prob = stats.poisson.pmf(5, lam)
print("Poisson Probability:", poisson_prob)
```

```
Lambda: 2.8545454545454545
Poisson Probability: 0.09094738770519475
```

[8]:
```python
# Log-Normal
amount_clean = amount[amount > 0]

shape, loc, scale = stats.lognorm.fit(amount_clean)
print("Log-Normal Parameters:", shape, loc, scale)

sns.histplot(amount_clean, kde=True)
plt.title("Log-Normal Distribution")
plt.show()
```

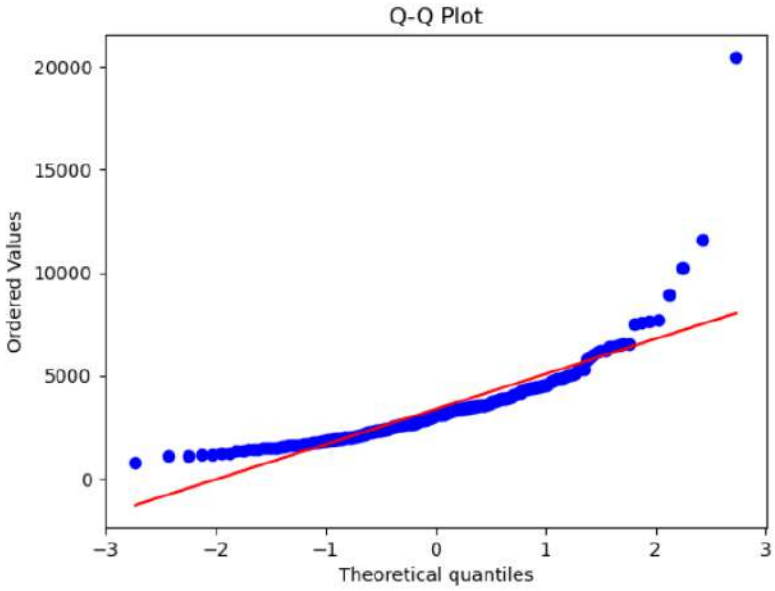
```
Log-Normal Parameters: 0.5416831487801056 333.7722748609052 2604.740667049811
```



[9]:
```
# Power Law

pareto_params = stats.pareto.fit(amount_clean)
print("Power Law Parameters:", pareto_params)
```

```
Power Law Parameters: (np.float64(107341796.39201096), np.float64(-274877906139.58), np.float64(274877906943.99997))
```

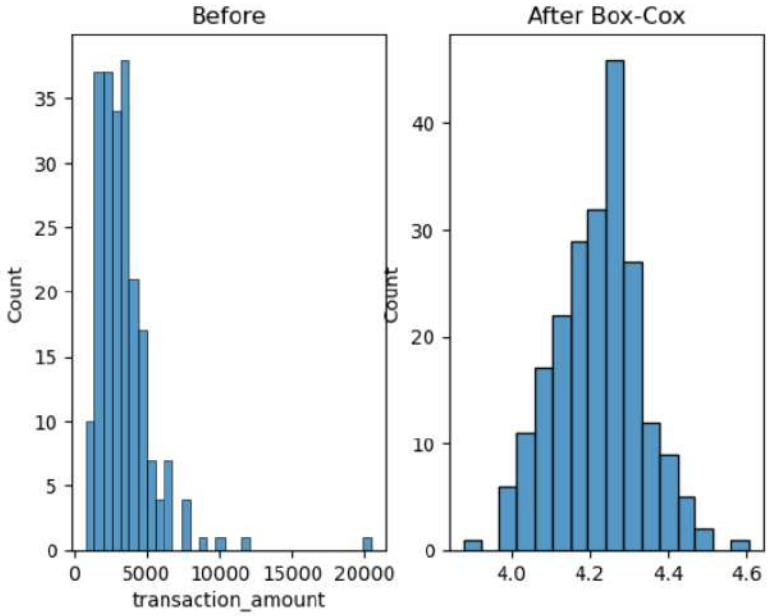
[10]:
```
# Q-Q Plot

stats.probplot(amount_clean, dist="norm", plot=plt)
plt.title("Q-Q Plot")
plt.show()
```



[11]:
```
# Box-Cox

amount_bc = amount_clean + 1

transformed, lam = stats.boxcox(amount_bc)
```

```python
print("Lambda:", lam)

plt.subplot(1,2,1)
sns.histplot(amount_clean)
plt.title("Before")

plt.subplot(1,2,2)
sns.histplot(transformed)
plt.title("After Box-Cox")

plt.show()
```

```
Lambda: -0.18123617450580254
```



[12]:
```
# Z-score

z_scores = stats.zscore(amount_clean)

df = df.loc[amount_clean.index]
df['Z_score'] = z_scores

print(df[['Z_score']].head())
```

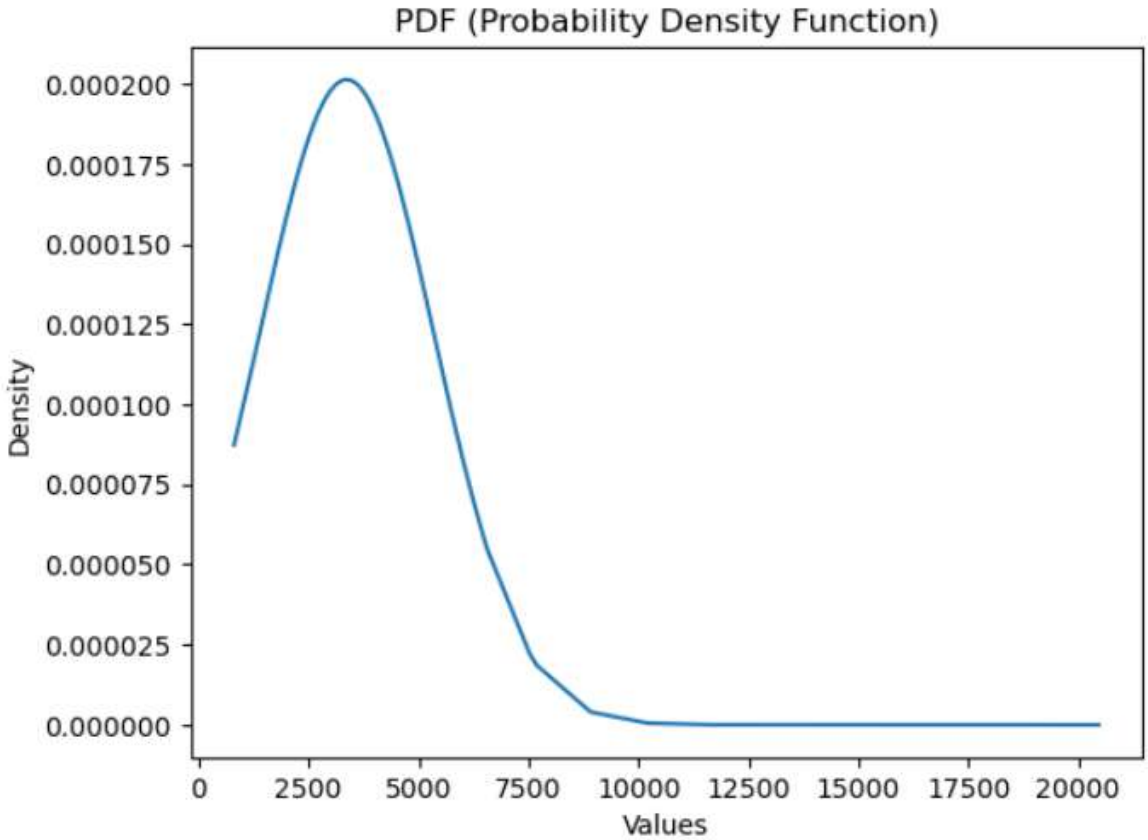
```
    Z_score
0  0.230240
1 -0.294446
2  0.381477
3  1.523626
4 -0.360179
```

[13]:
```
# Probability

prob = np.mean(amount_clean > 5000)
print("Probability > 5000:", prob)
```

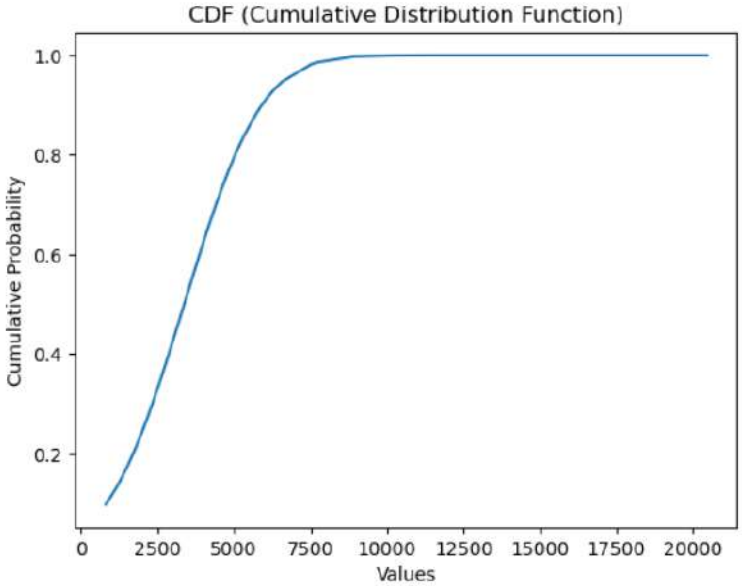
```
Probability > 5000: 0.11363636363636363
```

[14]:

```
# PDF & CDF

sorted_data = np.sort(amount_clean)

mean = np.mean(amount_clean)
std = np.std(amount_clean)

pdf = stats.norm.pdf(sorted_data, mean, std)

plt.plot(sorted_data, pdf)
plt.title("PDF (Probability Density Function)")
plt.xlabel("Values")
plt.ylabel("Density")
plt.show()

print("Explanation:")
print("The probability shows how likely a specific value is")

#CDF
cdf = stats.norm.cdf(sorted_data, mean, std)

plt.plot(sorted_data, cdf)
plt.title("CDF (Cumulative Distribution Function)")
plt.xlabel("Values")
plt.ylabel("Cumulative Probability")
plt.show()

print("Explanation:")
print("The CDF shows what the total probability is up to the given value.")
```




```
Explanation:
The probability shows how likely a specific value is
```




```
Explanation:
The CDF shows what the total probability is up to the given value.
```

[15]:
```python
# Conclusion
print("Conclusion:")
print("Check graphs and Q-Q plot to see which distribution fits best.")
print("If data is skewed → Log-Normal or Power Law is better.")
print("If data is symmetric → Normal distribution is better.")
```

```
Conclusion:
Check graphs and Q-Q plot to see which distribution fits best.
If data is skewed → Log-Normal or Power Law is better.
If data is symmetric → Normal distribution is better.
```